

Modeling Electromagnetic Fields

EE4375 - FEM For EE Applications

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Overview

- Electric Fields
- Magnetic Fields
- Thermal Fields (thermal losses)
- Mechanical Fields (deformation)
- Acoustic Fields (noise, vibration)

Stationary Electric Fields (1/4)

Nomenclature for Stationary Electric Field

(requires physical units to be filled in)

- ϵ_0 : permittivity of vacuum (scalar)
- $\epsilon(\mathbf{x})$: permittivity of material (scalar field)
- $\epsilon_r(\mathbf{x}) = \epsilon(\mathbf{x})/\epsilon_0$: relative permittivity (scalar field)
- $\rho(\mathbf{x})$: electrical charge density (scalar field)
- $\phi(\mathbf{x})$: electrical potential (scalar field)
- $\mathbf{E}(\mathbf{x})$: electric field (vector field)
- $\mathbf{D}(\mathbf{x})$: electric displacement field (vector field)

Stationary Electric Fields (2/4)

Maxwell Equation for Stationary Electric Field

- 1 $\nabla \times \mathbf{E} = \text{curl } \mathbf{E} = \mathbf{0}$ (vector-valued equation)
- 2 $\nabla \cdot \mathbf{D} = \text{div } \mathbf{D} = \rho$ (scalar equation)
- 3 $\mathbf{D} = \epsilon \mathbf{E}$ (vector-valued equation)

Stationary Electric Fields (3/4)

Maxwell Equation for Stationary Electric Field

- 1 From (1) follows that $\mathbf{E}$ is conservative, thus a potential ϕ exists such that $\mathbf{E} = -\nabla\phi = -\text{grad } \phi$ (minus sign follows convention) (vector-valued equation)
- 2 From (3) then follows that $\mathbf{D} = \epsilon \mathbf{E} = -\epsilon \nabla\phi$ (vector-valued equation)
- 3 From (2) then follows that $\nabla \cdot \mathbf{D} = -\nabla \cdot (\epsilon \nabla\phi) = \rho$ (scalar equation)

Stationary Electric Fields (4/4)

Poisson Equation for the Electric Potential

- case that $\epsilon = \epsilon(\mathbf{x})$ is spatially dependent

$$-\nabla \cdot (\epsilon \nabla \phi) = -\operatorname{div} (\epsilon \operatorname{grad} \phi) = \rho$$

- case that ϵ is constant (independent of $\mathbf{x}$)

$$-\nabla \cdot (\nabla \phi) = -\nabla^2 \phi = -\Delta \phi = \rho$$

- Laplace equation in case that $\rho = 0$

Stationary Magnetic Fields (1/)

Nomenclature for Stationary Magnetic Field

(requires physical units to be filled in)

- μ_0 : permeability of vacuum (scalar)
- $\mu(\mathbf{x})$: permeability of material (scalar field)
- $\mu_r(\mathbf{x}) = \mu(\mathbf{x})/\mu_0$: relative permeability (scalar field)
- $\mathbf{J}(\mathbf{x})$: applied current density (vector field)
- $\mathbf{M}(\mathbf{x})$: magnetization (vector field)
- $V_m(\mathbf{x})$: magnetic scalar potential (scalar field)
- $\mathbf{A}(\mathbf{x})$: magnetic vector potential (vector field)
- $\mathbf{B}(\mathbf{x})$: magnetic flux (vector field)
- $\mathbf{H}(\mathbf{x})$: magnetic field (vector field)

Stationary Magnetic Fields (2/)

Maxwell Equation for Stationary Magnetic Field

- 1 $\nabla \times \mathbf{H} = \text{curl } \mathbf{H} = \mathbf{J}$ (vector-valued equation)
- 2 $\nabla \cdot \mathbf{B} = \text{div } \mathbf{B} = 0$ (scalar equation)
- 3 $\mathbf{B} = \mu \mathbf{H}$ (vector-valued equation)

Stationary Magnetic Fields (3/)

Maxwell Equation for Stationary Magnetic Field

- 1 From (2) follows that a vector potential $\mathbf{A}$ exists such that $\mathbf{B} = \nabla \times \mathbf{A}$ (vector-valued equation)
- 2 From (3) then follows that $\mathbf{H} = 1/\mu \mathbf{B} = 1/\mu \nabla \times \mathbf{A}$ (vector-valued equation)
- 3 From (1) then follows that $\nabla \times \mathbf{H} = \nabla \times (1/\mu \nabla \times \mathbf{A}) = \mathbf{J}$ (vector equation)

Stationary Magnetic Fields (4/)

Double Curl Equation for the Magnetic Vector Potential

$$\textcircled{1} \quad \nabla \times \mathbf{H} = \nabla \times (1/\mu \nabla \times \mathbf{A}) = \mathbf{J}$$

Stationary Magnetic Fields (5/)

Two-Dimensional Perpendicular Current Formulation

- assume current perpendicular to 2D modeling plate
 $\mathbf{J} = (0, 0, J_z(x, y))$
- then $\mathbf{B} = (B_x(x, y), B_y(x, y), 0)$ and $\mathbf{H} = (H_x(x, y), H_y(x, y), 0)$
- then $\mathbf{A} = (0, 0, A_z(x, y))$
- $\frac{\partial}{\partial x} \left(\frac{1}{\mu} \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{1}{\mu} \frac{\partial A_z}{\partial y} \right) = J_z(x, y)$
- in post-processing $B_x(x, y) = -\frac{\partial A_z}{\partial y}$, $H_x = 1/\mu B_x$, $B_y(x, y) = \frac{\partial A_z}{\partial x}$,
 $H_y = 1/\mu B_y$

Stationary Magnetic Fields (6/)

Models for Magnetic Saturation

Transient Quasi-Stationary Magnetic Fields (1/2)

Nomenclature for Transient Quasi-Stationary Magnetic Field

(requires physical units to be filled in)

- ι : imaginary unit
- σ : electrical conductivity (scalar)
- $\omega = 2\pi f$: frequency

Transient Quasi-Stationary Magnetic Fields (2/2)

- assume contribution from displacement current can be neglected
- partial differential equation for the magnetic vector potential $\mathbf{A}$

$$\sigma \frac{\partial \mathbf{A}}{\partial t} + \nabla \times (1/\mu \nabla \times \mathbf{A}) = \mathbf{J}$$

Time-Harmonic Quasi-Stationary Magn Fields (1/2)

- assume time-harmonic signals at frequency $\omega = 2\pi f$
- partial differential equation for the magnetic vector potential **A**
phasor $\iota\omega\sigma\mathbf{A} + \nabla \times (1/\mu\nabla \times \mathbf{A}) = \mathbf{J}$
- perpendicular current formulation
$$\iota\omega\sigma A_z + \frac{\partial}{\partial x} \left(\frac{1}{\mu} \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{1}{\mu} \frac{\partial A_z}{\partial y} \right) = J_z(x, y)$$
- assumption of linearity (superposition) violated in the presence of non-linear effects

Time-Harmonic Quasi-Stationary Magn Fields (2/2)

Models for Skin Effect

Electrical Conductor Models (1/2)

Stranded Conductor

- current-driven: no circuit equation
- voltage-driven: sum of resistive and induced voltage is equal to applied voltage

Electrical Conductor Models (2/2)

Solid Conductor

- current-driven: sum of resistive and induced current is equal to applied current
- voltage-driven: no circuit equation

Field-Circuit Coupled Models (1/1)

- magnetic field: PDE for magnetic (scalar or vector) potential
- electrical circuit: ODE for applied voltage and current on conductors
- coupling: induced voltages in stranded conductors and induced currents in solid conductors

Electro-Magnetic Loses (1/2)

- copper losses in the winding - Ohm's Law - Losses proportional to I^2 ;

Electro-Magnetic Loses (2/2)

- iron losses in the core - Steinmetz's formula - material properties;

Thermal Fields

- Poisson equation for the thermal field

Mechanical Fields

- deformation, strain, stress
- Navier-Stokes equations for a linear elastic medium

Acoustical Fields

- wave-equation
- Helmholtz-equation

References

- list references here